100% VERIFIED HARDWARE & BOOTLOADER GUIDE • ENGINEERED BY 0XMEHR4B

POCO X3 Pro (vayu / bhima) Ultimate Bootloader Unlocker

Permanent Bootloader Unlock for Error 20091 / 20031 & Reballed/Unfused SM8150 Devices via Zero-Auth EDL (9008) • **by 0xmehr4b**

1. Why Official Mi Unlock Fails with Error 20091 & 20031

Many POCO X3 Pro (`vayu` / `bhima` , Snapdragon 860 / `SM8150`) devices have had their SoC reballed or replaced with an unfused `SM8150` chip (reporting `secure: no` in Fastboot).

- **Why Settings → Mi Unlock Status returns `20091` :** `Settings.apk` (`CloudDeviceStatus.bindAccountWithDevice`) asks Qualcomm TEE (`cardapp`) to sign the bind request using the factory FID key. On an unfused replacement CPU, `cardapp` fails with `Error 41` , Xiaomi's server responds `No available key on server` , and `Settings.apk` hardcodes `error_code = 20091` .
- **Why EDL 9008 Needs Zero Xiaomi Auth on `secure: no` :** Disassembly of `prog_ufs_firehose_sm7150_ddr.elf` (at `0x14839868 → 0x1483d904`) confirms that the programmer checks `boot_is_auth_enabled()` . When `secure: no` , it returns `0` and **completely bypasses Xiaomi's EDL `<sig>` authentication**.
- **Why Patching `devinfo` Permanently Unlocks `abl.elf` :** At boot (`0x2b4e4`), `abl.elf` checks `IsSecureBootEnabled()` ; when `0` , it skips `oem_misc1` token checks and trusts `devinfo` byte `0x0D` (`is_unlocked = 1`) and byte `0x0E` (`is_unlock_critical = 1`).

2. Package Contents & GUI Overview

Double-click `One_Click_Unlock_POCO_X3_Pro.bat` or `Run_Unlocker_GUI.bat` to launch the unlocker.

POCO X3 Pro (vayu / bhima) Ultimate Bootloader Unlocker v1.0 • by 0xmehr4b

Zero-Auth EDL 9008 UFS devinfo Unlocker • Solves Mi Unlock Error 20091 / 20031 •

Engineered by 0xmehr4b

`EDL 9008 READY (COM3)`

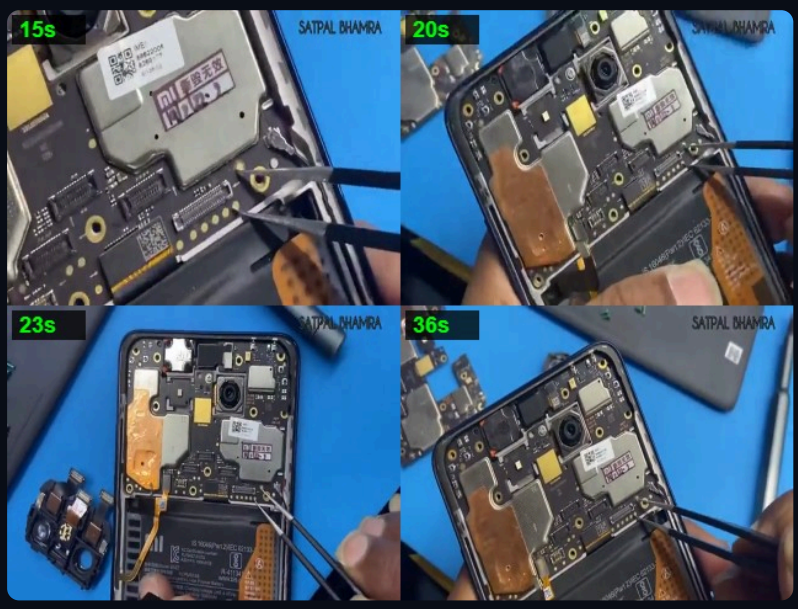
[1-CLICK] START AUTO-UNLOCK (Wait for EDL 9008 & Unlock Automatically)

```
[+] Connected to Qualcomm HS-USB QDLoader 9008 on COM3 (\\.\COM3)
[+] Stage 2/6: Uploaded prog_ufs_firehose_sm7150_ddr.elf (741,180 bytes in 0.094s)
[+] Stage 3/6: Backed up original devinfo → backups/backup_devinfo_YYYYMMDD_HHMMSS.bin
[+] Stage 4/6: Patched unlock flags (is_unlocked=1, is_unlock_critical=1)
[+] Stage 5/6: VERIFIED 100% BYTE-FOR-BYTE on UFS LUN 0 Sector 12288!
[+] Stage 6/6: Reboot command sent. Bootloader is now UNLOCKED!
```

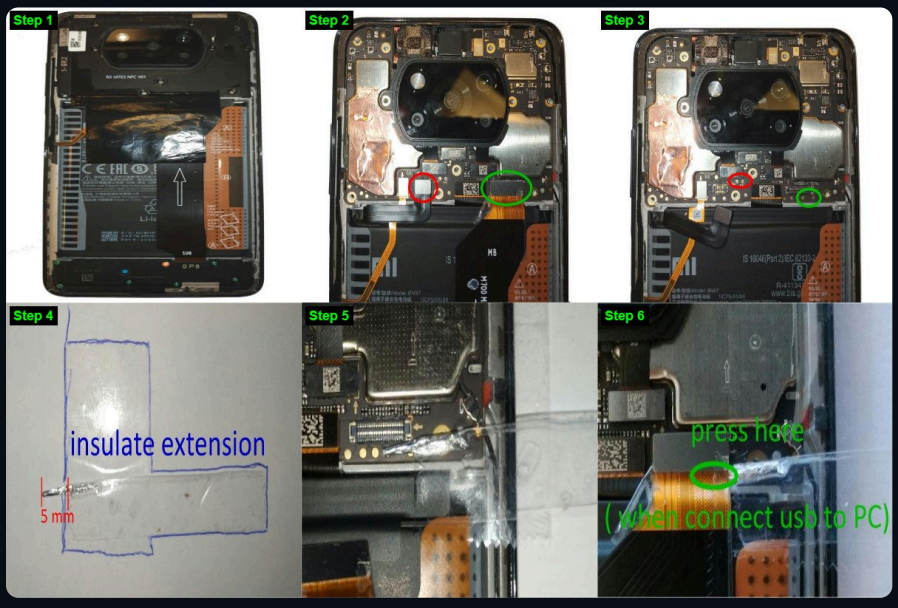
File / Folder	Purpose
`One_Click_Unlock_POCO_X3_Pro.bat`	All-in-One Runner: Auto-checks/installs drivers & unlocks in EDL 9008
`Run_Unlocker_GUI.bat`	Launches the Dark-Mode GUI Unlocker with Auto-Wait & Live Monitor
`Run_Auto_Unlock_CLI.bat`	Launches CLI Auto-Wait mode (waits for 9008 and unlocks automatically)
`Install_Drivers.bat`	One-click Admin installer for Qualcomm 9008 + Fastboot WinUSB drivers
`bin/` & `firmware/`	Standalone `QSaharaServer.exe`, `fh_loader.exe`, `adb`, `fastboot`, and `vayu` Firehose
`backups/` & `logs/`	Automatic timestamped backups of original `devinfo` & session logs

3. Exact POCO X3 Pro (vayu) Hardware EDL Test Points

> **CRITICAL RULE FOR THE WIDE "MB" RIBBON CABLE:**
> In frame `15s` below, the wide black `MB` ribbon cable is unplugged *only* so the camera can clearly see the gold dots. **You MUST keep the wide `MB` ribbon cable PLUGGED IN** (as shown in frames `20s` and `36s`) when shorting the two points and connecting the USB cable, because the `MB` cable carries the USB-C connection from the charging port to the motherboard!



Verified POCO X3 Pro (vayu) EDL Test Points: Point 1 (Top Tweezer Tip): Single gold dot just above the right corner of the wide `MB` connector (under the silver shield corner). **Point 2 (Bottom Tweezer Tip):** The 5th (rightmost) gold dot in the row of 5 dots directly below the `MB` connector.



Teardown Overview: Remove back cover & upper plastic shield screws, unplug the Battery connector (red circle in Step 2), and leave the wide `MB` USB ribbon connected when bridging the two right-edge EDL test points.

4. Step-by-Step Bootloader Unlock Procedure

STEP 1

Install Drivers & Arm Auto-Unlock

Run `One_Click_Unlock_POCO_X3_Pro.bat` (which auto-installs drivers and arms Auto-Unlock automatically), or open `Run_Unlocker_GUI.bat` and click **"[1-CLICK] START AUTO-UNLOCK"**.

STEP 2

Disconnect Battery & Short EDL Points

Unplug the USB cable and disconnect the **Battery connector**. Keep the wide **MB cable connected**. Hold metal tweezers firmly across **Point 1** (upper-right dot) and **Point 2** (5th lower-right dot).

STEP 3

Plug In USB & Automatic Flash

While holding the tweezers on both dots, plug the USB-C cable into the phone. After 2 seconds (Windows USB chime), release the tweezers and click the **Battery connector** firmly back into place.

STEP 4

Automatic Backup, Unlock & Verify

Within 2 seconds, the tool uploads the Firehose programmer, backs up your original `devinfo` to `backups/`, writes `is_unlocked=1`, verifies it byte-for-byte, and reboots the phone!

5. Post-Unlock First Boot & Troubleshooting Guide

1. Phone Shows Charging Icon / Red Bar After EDL Reset:
If the phone shows an empty battery/plug icon with a red bar underneath after EDL reset, the **Battery connector** is not fully clicked down. Unplug USB, press the Battery connector firmly onto the motherboard socket until it clicks, then hold the **Power button for 5 seconds**. You will see the **POCO logo with the Unlocked padlock icon at the top**!

2. Phone Enters Xiaomi Default Recovery Mode After Unlock:
On Android 12+, Qualcomm Keymaster (`km41`) binds the `/data` encryption key to the bootloader lock state (`device_locked`). When transitioning from Locked (`1`) to Unlocked (`0`), you must format `/data` once (just like official Mi Unlock does):
In Xiaomi Recovery, select **Wipe Data** → **Wipe All Data** → **Confirm**, then select **Reboot to System**.

3. Bypassing Xiaomi Account Email 2FA "Too Many Attempts, Try Again Tomorrow":
If the phone had a Xiaomi Account signed in before wiping data and asks for Email 2FA during initial setup, but Xiaomi says "*Too many attempts, try again tomorrow*":
• Open your already-logged-in Xiaomi Account session at `account.xiaomi.com` on your PC browser.
• Change the account's linked **Email Address** to a secondary email.
• Xiaomi immediately resets the 2FA email counter for the new email address—verify the code, sign in on the phone, and remove the Xiaomi Account cleanly!
• **Important:** Never use `flash_all_lock.bat` ("clean all and lock" in MiFlash) or `fastboot oem lock` in the future, so your bootloader remains permanently unlocked.

Copyright © 2026 **Mohammad Mehrab Hossain (0xmehr4b)** • github.com/0xmehr4b • All rights reserved.